



HOMEWORK REPORT

A Descriptive Report Title

DDA5001 Machine Learning | Student Name | Student ID | September 29, 2026

Abstract

Template note. Replace this paragraph with a concise abstract only when appropriate. State the task, approach, and verified result without claiming work that has not been performed.

1 Introduction

Use this section to state the question and the scope of the report. Replace all illustrative text before submission. For a problem-set assignment, sections may instead be titled “Problem 1,” “Problem 2,” and so on.

1.1 Mathematical notation

Number important displayed equations and explain each symbol in the surrounding prose. For example, a generic linear model can be written as

$$y = X\theta + \varepsilon, \quad (1)$$

where $X \in \mathbb{R}^{n \times d}$, $\theta \in \mathbb{R}^d$, and $y, \varepsilon \in \mathbb{R}^n$. Refer to Equation (1) by its label.

2 Methods

Describe assumptions, algorithms, parameters, and data sources precisely enough for another reader to reproduce the work. Use a short listing only when code is needed to clarify a method; submit complete source files separately if required.

```
1 def predict(X, theta):  
2     return X @ theta
```

Listing 1. Illustrative Python syntax; replace with code actually used.

3 Results and Discussion

Insert only results supported by saved program output or a documented derivation. Table 1 demonstrates the table style; it does not report an experiment.

Table 1. Notation used in the illustrative linear model.

Symbol	Meaning
n	Number of observations
d	Number of features
X	Design matrix in $\mathbb{R}^{n \times d}$